

SECOND SEMESTER DIPLOMA EXAMINATION IN ENGINEERING
AND TECHNOLOGY

BUSINESS STATISTICS
MODEL QUESTION PAPER – SET-1

Time: 3 hours

Maximum Marks: 75

PART A

I. Answer all questions in one word or one sentence. Each question carries one mark.

(9 x 1 = 9 Marks)

1	In statistics, _____ classification includes data according to the time period in which the items under consideration occurred. a) Chronological b) Alphabetical c) Geographical d) Topological	M1.8	R
2	When an investigator uses the data which has already been collected by others, such data is called _____	M1.05	R
3	The value lying half way between the upper limit and lower limit of the class is:	M2.01	R
4	The number of observations in a particular class is called:	M2.02	R
5	Correlation value r lies between –1 and	M3.01	R
6	Type of correlation which Studies more than two variables	M3.02	R
7	Line which gives the best estimate of dependent variable for any given value of independent variable.	M3.01	R
8	-----are specialized averages designed to measure the change in a group of related variables over a period of time.	M4.01	R
9	A test to determine whether a given method will work both ways in time, forward and back ward.	M4.04	R

PART B

II. Answer any eight questions from the following. Each question carries 3 marks

(8 x 3 = 24 Marks)

1	State the functions of statistics.	M1.01	U
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2	Find combined mean for the following data. Group Number Mean A 200 25 B 250 10 C 300 15	M2.05	U
3	State the meaning and importance of correlation.	M3.05	U
4	List the Characteristics of Index number.	M4.06	R
5	State the merits of median.	M2.03	U
6	Find out mode form the following. Serial No 1 2 3 4 5 6 7 8 9 10 Marks obtained 60 77 74 62 77 77 70 68 65 80	M2.03	U
7	Distinguish between primary data and secondary data.	M1.05	A
8	State the importance of measuring variation or dispersion.	M2.06	U
9	State the limitations of index numbers.	M4.01	U
10	Write short note on in individual series, discrete series and continuous series.	M1.04	R

PART C

Answer all questions. Each question carries seven marks

(6 x 7 = 42 Marks)

III	Explain the rules of tabulation.	M1.09	U											
OR														
IV	What are the various kinds of classification?	M1.08	R											
V	Calculate mean deviation about median and co-efficient of mean deviation. 0-10 10-20 20-30 30-40 40-50 50-60 60-70 70-80 18 16 15 12 10 5 2 2	M2.06	A											
OR														
VI	Compute Standard Deviation form the following data of the income of 10 employees of a firm. <table border="1"><tr><td>Monthly income</td><td>600</td><td>620</td><td>640</td><td>620</td><td>680</td><td>670</td><td>680</td><td>640</td><td>700</td><td>650</td></tr></table>	Monthly income	600	620	640	620	680	670	680	640	700	650	M2.07	U
Monthly income	600	620	640	620	680	670	680	640	700	650				
VII	Compute correlation coefficient. Price index (X) 120 150 190 220 230 Money supply in Rs crores(Y) 1800 2000 2500 2700 3000	M3.04	U											
OR														

VIII	Obtain Spearman's rank correlation. Individuals : A B C D E F G H I J Rank before : 1 6 3 9 5 2 7 10 8 4 Rank after : 6 8 3 2 7 10 5 9 4 1	M3.04	U																								
IX	Explain the meaning and purposes of index numbers.	M4.01	U																								
OR																											
X	Calculate cost of living index number from the following data. <table><tr><td>Items</td><td>Weight</td><td>Price in Base year</td><td>Price in Current year</td></tr><tr><td>Food</td><td>10</td><td>150</td><td>225</td></tr><tr><td>House Rent</td><td>5</td><td>50</td><td>150</td></tr><tr><td>Clothing</td><td>2</td><td>30</td><td>60</td></tr><tr><td>Fuel</td><td>3</td><td>30</td><td>75</td></tr><tr><td>Others</td><td>5</td><td>50</td><td>75</td></tr></table>	Items	Weight	Price in Base year	Price in Current year	Food	10	150	225	House Rent	5	50	150	Clothing	2	30	60	Fuel	3	30	75	Others	5	50	75	M4.04	U
Items	Weight	Price in Base year	Price in Current year																								
Food	10	150	225																								
House Rent	5	50	150																								
Clothing	2	30	60																								
Fuel	3	30	75																								
Others	5	50	75																								
XI	Find mean from the following data. <table><tr><td>Roll No</td><td>Marks</td></tr><tr><td>1.</td><td>21</td></tr><tr><td>2.</td><td>30</td></tr><tr><td>3.</td><td>28</td></tr><tr><td>4.</td><td>40</td></tr><tr><td>5.</td><td>26</td></tr><tr><td>6.</td><td>34</td></tr><tr><td>7.</td><td>40</td></tr><tr><td>8.</td><td>9</td></tr><tr><td>9.</td><td>15</td></tr><tr><td>10.</td><td>17</td></tr></table>	Roll No	Marks	1.	21	2.	30	3.	28	4.	40	5.	26	6.	34	7.	40	8.	9	9.	15	10.	17	M2.04	U		
Roll No	Marks																										
1.	21																										
2.	30																										
3.	28																										
4.	40																										
5.	26																										
6.	34																										
7.	40																										
8.	9																										
9.	15																										
10.	17																										
OR																											
XII	State the merits and demerits of mean deviation.	M2.06	U																								
XIII	State the importance of index numbers.	M4.02	U																								
OR																											
XIV	Calculate simple aggregative price index. <table><tr><td>Commodity</td><td>Base Period Price (rs)</td><td>Current Period Price (rs)</td><td>Percentage Change</td></tr><tr><td>A</td><td>2</td><td>4</td><td>100</td></tr><tr><td>B</td><td>5</td><td>6</td><td>20</td></tr><tr><td>C</td><td>4</td><td>5</td><td>25</td></tr><tr><td>D</td><td>2</td><td>3</td><td>50</td></tr></table>	Commodity	Base Period Price (rs)	Current Period Price (rs)	Percentage Change	A	2	4	100	B	5	6	20	C	4	5	25	D	2	3	50	M4.03	U				
Commodity	Base Period Price (rs)	Current Period Price (rs)	Percentage Change																								
A	2	4	100																								
B	5	6	20																								
C	4	5	25																								
D	2	3	50																								

SECOND SEMESTER DIPLOMA EXAMINATION IN ENGINEERING
AND TECHNOLOGY

BUSINESS STATISTICS
MODEL QUESTION PAPER – SET-2

Time: 3 hours

Maximum Marks: 75

PART A

I. Answer all questions in one word or one sentence. Each question carries one mark.

(9 x 1 = 9 Marks)

1	The most commonly used device of presenting business and economic data is: a) Pie diagrams b) Pictograms c) Bar diagrams d) Line diagrams	M1.10	R
2	A systematic arrangement of data in rows and columns is	M1.08	R
3	The classes in which the lower limit or the upper limit is not specified are known as:	M2.01	R
4	For a frequency distribution mean deviation from mean is computed by a) $\sum E f / \sum E f d $ b) $\sum E d / \sum E f$ c) $\sum E f d / \sum E f$ d) $\sum E f d / \sum E f$	M2.07	R
5	-----analysis studies the relation between two variables	M3.01	R
6	Who coined the concept of rank correlation	M3.03	R
7	Who first used the term regression	M3.05	R
8	-----year with which the changes in the current year are compared for indexation.	M4.01	R
9	Which test shows an index of price, if multiplied by an index of quantity, with the same base, given years and commodities, should be equal to the true value ratio	M4.04	R

PART B

II. Answer any eight questions from the following. Each question carries 3 marks

(8 x 3 = 24 Marks)

1	What is frequency polygon?	M1.10	R
2	How will you calculate median for continuous series?	M2.01	R
3	Describe two Regression Lines.	M3.05	U
4	What is Family Budget Method	M4.06	R
5	What are the essentials of an Ideal Average?	M2.03	R
6	State the Merits of Median.	M2.03	U
7	List out the Parts of tables.	M1.09	R
8	What are the Characteristics of a Measure of Variation?	M2.06	R
9	What is Consumer price Index Number (or) Cost of living index?	M4.04	R
10	State the characteristics of Statistics.	M1.01	U

PART C

Answer all questions. Each question carries seven marks

(6 x 7 = 42 Marks)

III.	Explain the methods of collecting primary data	M1.04	U																			
OR																						
IV.	State the limitations of statistics.	M1.03	U																			
V.	Following data relates to daily wages of persons working in a factory. Compute the median daily wage.		M2.02	A																		
	<table><tr><td>Daily wages (in Rs)</td><td>Number of workers</td></tr><tr><td>55–60</td><td>7</td></tr><tr><td>50–55</td><td>13</td></tr><tr><td>45–50</td><td>15</td></tr><tr><td>40–45</td><td>20</td></tr><tr><td>35–40</td><td>30</td></tr><tr><td>30–35</td><td>33</td></tr><tr><td>25–30</td><td>28</td></tr><tr><td>20–25</td><td>14</td></tr></table>				Daily wages (in Rs)	Number of workers	55–60	7	50–55	13	45–50	15	40–45	20	35–40	30	30–35	33	25–30	28	20–25	14
	Daily wages (in Rs)	Number of workers																				
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25–30	28																					
20–25	14																					
OR																						
VI	List the merits and demerits of standard deviation.	M2.07	R																			

VII	Calculate Karl Pearson's Correlation for the following data. X: 2 3 4 5 6 7 8 Y: 4 5 6 12 9 5 4	M3.04	U																					
OR																								
VIII	Explain types of correlation.	M3.03	U																					
IX	Construct the price index number for 2003, taking the year 2000 as base year	M4.03	A																					
	<table><tr><td rowspan="2">Commodity</td><td>Price in the year</td><td>Price in the year</td></tr><tr><td>2000</td><td>2003</td></tr><tr><td>A</td><td>60</td><td>80</td></tr><tr><td>B</td><td>50</td><td>60</td></tr><tr><td>C</td><td>70</td><td>100</td></tr><tr><td>D</td><td>120</td><td>160</td></tr><tr><td>E</td><td>100</td><td>150</td></tr></table>			Commodity	Price in the year	Price in the year	2000	2003	A	60	80	B	50	60	C	70	100	D	120	160	E	100	150	
	Commodity				Price in the year	Price in the year																		
				2000	2003																			
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X	State the importance of index numbers.	M4.01	U																					
XI	Find mean, from the following data. 0-10 10-20 20-30 30-40 40-50 50-60 60-70 3 10 15 20 12 7 3	M2.04	U																					
OR																								
XII	Calculate mean deviation and its coefficient from arithmetic mean for the following the data 100 150 200 250 300	M2.08	U																					
XIII	Given are the following price-quantity data of fish, with price quoted in Rs per kg and production in quintals.	M4.03	A																					
	<table><tr><td>Year</td><td>Price</td><td>Production</td></tr><tr><td>1980</td><td>15</td><td>500</td></tr><tr><td>1981</td><td>17</td><td>550</td></tr><tr><td>1982</td><td>16</td><td>480</td></tr><tr><td>1983</td><td>18</td><td>610</td></tr><tr><td>1984</td><td>22</td><td>650</td></tr><tr><td>1985</td><td>20</td><td>600</td></tr></table>			Year	Price	Production	1980	15	500	1981	17	550	1982	16	480	1983	18	610	1984	22	650	1985	20	600
	Year			Price	Production																			
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	1983			18	610																			
1984	22	650																						
1985	20	600																						
Construct: (a) The price index for each year taking price of 1980 as base.																								
OR																								
XIV	Explain the various tests of adequacy of index number.	M4.04	U																					

SCHEME OF VALUATION FOR QUESTION PAPER 1

BUSINESS STATISTICS

Qn No.	Scoring Indicator	Split up Score	Total
I	<u>PART : A</u>		9x1=9
1	a) Chronological	1	9
2	Secondary data	1	
3	midpoint	1	
4	Frequency	1	
5	+1	1	
6	Multiple correlation	1	
7	Regression line	1	
8	Index numbers	1	
9	Time reversal test	1	
II	<u>PART B</u>		3x8 = 24
1	• Simplifies complexity of data • presents facts in a definite and precise form • condensation • comparison of data • testing • prediction • Policy formulation (6 points *0.5=3)	3	
2	$\frac{N_1\bar{X}+N_2\bar{X}^2}{N_1+N_2}$ ANSWER =16 (equation and steps 2 mark final answer 1 mark)	3	
3	Correlation is concerned with describing the degree of relation between two variables Importance of Correlation (i) Correlation helps us in determining the degree of relationship between variables. It enables us to make our decision for the future course of actions. (ii) Correlation analysis helps us in understanding the nature and degree of relationship which can be used for future planning and forecasting. (iii) Forecasting without any prior correlation analysis may prove to be defective, less reliable and more uncertain. If it is based upon the result of correlation analysis, it will be more reliable. (meaning 1 mark at least 2 importance 2 mark)	3	
4	They are specialized average - They measure the net change in a group of related variables - They measure the effect of changes over a period of time -They help comparison of groups of variables directly- a Ratio of two numbers expressed as percent-measuring difference in	3	

	the magnitude of a group of related variables (3points *1=3 mark)																																
5	➤ It is easy to compute and understand ➤ It eliminates the effect of extreme item ➤ The value of median can be located graphically 3 points *1 mark	3																															
6	Marks obtained by 10 students is here 77 is repeated three times Therefore the Mode mark is 77 Basic steps 2 mark final answer 1 mark	3																															
7	<table><tr><th>BASIS</th><th>PRIMARY DATA</th><th>SECONDARY DATA</th></tr><tr><td>Meaning</td><td>Primary data refers to the first hand data gathered by the researcher himself.</td><td>Secondary data means data collected by someone else earlier.</td></tr><tr><td>Data</td><td>Real time data</td><td>Past data</td></tr><tr><td>Process</td><td>Very involved</td><td>Quick and easy</td></tr><tr><td>Source</td><td>Surveys, observations, experiments, questionnaire, personal interview, etc.</td><td>Government publications, websites, books, journal articles, internal records etc.</td></tr><tr><td>Cost effectiveness</td><td>Expensive</td><td>Economical</td></tr><tr><td>Collection time</td><td>Long</td><td>Short</td></tr><tr><td>Specific</td><td>Always specific to the researcher's needs.</td><td>May or may not be specific to the researcher's need.</td></tr><tr><td>Available in</td><td>Crude form</td><td>Refined form</td></tr><tr><td colspan="3">3 difference *1</td></tr></table>	BASIS	PRIMARY DATA	SECONDARY DATA	Meaning	Primary data refers to the first hand data gathered by the researcher himself.	Secondary data means data collected by someone else earlier.	Data	Real time data	Past data	Process	Very involved	Quick and easy	Source	Surveys, observations, experiments, questionnaire, personal interview, etc.	Government publications, websites, books, journal articles, internal records etc.	Cost effectiveness	Expensive	Economical	Collection time	Long	Short	Specific	Always specific to the researcher's needs.	May or may not be specific to the researcher's need.	Available in	Crude form	Refined form	3 difference *1			3	
BASIS	PRIMARY DATA	SECONDARY DATA																															
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Specific	Always specific to the researcher's needs.	May or may not be specific to the researcher's need.																															
Available in	Crude form	Refined form																															
3 difference *1																																	
8	➤ Testing the Reliability of the Measures of Central Tendency ➤ Comparing two or more series on the basis of their variability ➤Enabling to control the variability ➤ Facilitating as a Basis for further statistical Analysis 3 points *1	3																															
9	If the chosen base year is not a normal one, the purpose is list ➤ Every index number has its own purpose. No index number can serve	3																															

	all purpose ➤ These are only appropriate indications of the relative level. 3 points *1		
10	Individual Series An individual series is one in which each value of the variable occurs for only once. In other words, the frequency of occurrence of all the values in such a series is only one. As such, essentially such series are displayed without the frequency column. The following are the examples of individual series. A discrete series is one in which the different values of a variable are shown in a discontinuous manner along with their respective frequencies and at least one of the values has a frequency of more than one. A continuous series is one which the different values of the variables are stated in a continuous manner along with their respective frequencies. Such series can be arrayed either in ascending, or in descending order. Each concept carries 1 mark	3	
	<u>PART C</u>		15*4 =30
<u>III</u>	(i) A good table must contain all the essential parts, such as, Table number, Title, Head note, Caption, Stub, Body, Foot note and source note. (ii) A good table should be simple to understand. It should also be compact, complete and self-explanatory. (iii) A good table should be of proper size. There should be proper space for rows and columns. One table should not be overloaded with details. Sometimes it is difficult to present entire data in a single table. In that case, data are to be divided into more number of tables. (iv) A good table must have an attractive get up. It should be prepared in such a manner that a scholar can understand the problem without any strain. (v) Rows and columns of a table must be numbered. (vi) In all tables the captions and stubs should be arranged in some systematic manner. The manner of presentation may be alphabetically, or chronologically depending upon the requirement. (vii) The unit of measurement should be mentioned in the head note. (viii) The figures should be rounded off to the nearest hundred, or thousand or lakh. It helps in avoiding unnecessary details (7 item*1=7,)	7	
<u>IV-</u>	<u>Bases of Classification</u> There are four important bases of classification: (1) Qualitative Base (2) Quantitative Base (3) Geographical Base (4) Chronological or Temporal Base (1) Qualitative Base	7	

	When the data are classified according to a quality or attribute such as sex, religion, literacy, intelligence, etc. (2) Quantitative Base When the data are classified by quantitative characteristics like height, weight, age, income, etc. (3) Geographical Base When the data are classified by geographical regions or location, like states, provinces, cities, countries, etc. (4) Chronological or Temporal Base When the data are classified or arranged by their time of occurrence, such as years, months, weeks, days, etc. (1) One -way Classification If we classify observed data keeping in view a single characteristic, this type of classification is known as one-way classification. (2) Two -way Classification If we consider two characteristics at a time in order to classify the observed data then we are doing two way classification. For example: The population of the world may be classified by religion and sex. (3) Multi -way Classification We may consider more than two characteristics at a time to classify given or observed data. In this way we deal in multi-way classification. (7 classification each *1)		
<u>V</u>	. Median=24 Mean deviation =14.775 Co-efficient =.616 (median 3 mark, mean deviation 3 mark, co-efficient-1 mark)	7	
<u>VI</u>	step 1 calculate X from $X = \frac{\Sigma X}{N}$-----2 mark Step 2 calculate X^2-----2 mark $\sigma = \sqrt{\Sigma X^2 / N}$-----1 mark $\sqrt{9200 / 10} = 30.3$-----2 mark $\sigma = 30.3$	7	
<u>VII</u>	The simplification, using step deviation method is illustrated below. Let A = 100; h = 10; B = 1700 and k = 100 $\Sigma U = 41$; $\Sigma U^2 = 35$; $\Sigma U^2 = 423$; $\Sigma V^2 = 343$; $\Sigma UV = 378$ R=0.98 Each steps carries 1 mark	7	

VIII	$\frac{1-6\sum D^2}{N^3-N}$ $1-1.07 = -0.07$ Equation=1 marks , steps =3 marks , final answer =3 marks	7	
IX	1. They help in framing suitable policies 2. They reveal trends and tendencies 3. Index numbers help us in forecasting future economic activity 4. Index numbers are very useful in deflating (any 4 explain x1.5Mark ,over all concept presentation 1 mark)	7	
X	$CPI = \frac{\sum WP}{\sum W} = \frac{4900}{25} = 196$ (Equation =2 Mark .steps 3 mark and final answer 2 Mark)	7	
XI	Formula = $X = \frac{\sum X}{N}$ -----1 mark $= \frac{300}{10} = 30$-----3 mark The mean marks =30-----3 mark	7	
XII	Merits : - It is simple to understand and easy to compute. - Based on each and every item of the data. - Least affected by the extreme values. - It can be computed from any average, mean, median or mode. Demerits : - Signs are ignored therefore mathematically it is incorrect and not a significant measure. - Cannot be compared if mean deviations of different series are based on different averages. - Does not give accurate results. 3 mark for merit 3 mark for demerit over all concept presentation 1 mark	7	
XIII	i. Help in Framing Suitable Policies (ii) They Reveal Trends (iii) Helps in Comparative Study (iv) Important in Forecasting (v) Universal Application (explain with suitable examples)	7	
XIV	<i>simple aggregative price index</i> is $P_{01} = \frac{\sum P_1}{\sum P_0} \times 100$-----3 mark $P_{01} = \frac{4+6+5+3}{2+5+4+2} \times 100$-----3 mark $= 138.5$-----1 mark	7	

SCHEME OF VALUATION FOR QUESTION PAPER 2
BUSINESS STATISTICS

Qn No.	Scoring Indicator	Split up Score	Total
I	<u>PART : A</u>		9x1=9
1	c) Bar diagrams	1	9
2	Table	1	
3	Open ended	1	
4	d) $\sum Ef d / \sum E f$	1	
5	Simple correlation (correlation)	1	
6	Spearman	1	
7	Francis Galton	1	
8	Base year	1	
9	Factor reversal test	1	
II	<u>PART B</u>		3x8 = 24
1	Frequency polygon is a line graph of class frequency plotted against class mid-point. It can be obtained by joining the midpoints of the tops of the rectangles in the histogram.	3	
2	• Arrange the variables in ascending or descending order. • Calculate cumulative frequencies. • Determine median class by using (N/2). • Apply formula – $M = l_1 + \frac{i}{f}(m - c)$ • l_1 □□ lower limit of median group • i □□ class interval • m □ N/2 • c □ cumulative frequency preceding the median group. f □□ frequency of median group 6 steps *0.5=3	3	
3	Regression line is that line which gives the best estimate of dependent variable for any given value of independent variable. If we take the case of two variables X and Y, we shall have two regression lines as the regression of X on Y and the regression of Y on X. Regression Line X and Y :In this formation, Y is independent and X is dependent variable, and best expected value of X is calculated corresponding to the given value of Y. Regression Line Y on X :Here Y is dependent and X is independent variable, best expected value of Y is estimated equivalent to the given value of X. (each regression line 1.5 mark)	3	
4	Family Budget Method :In this method, the price relatives for each commodity are obtained and these price relatives are multiplied by the value weights for each item and the product is divided by the total of weights.	3	

	Consumer price index = $\frac{\sum RW}{\sum W}$ Where R = $\frac{P_1}{P_0} \times 100$ and W = P₀ q₀ Meaning -1.5 mark equation 1.5 mark		
5	(i) Should be easy to understand. (ii) Clearly and rigidly defined. (iii) Based on all the observations. (iv) Simple to compute. (v) Least affected by fluctuations. (vi) Capable of further Algebraic treatment. (vii) Sampling stability. Points *0.5= 3 mark	3	
6	- It is easy to compute and understand - It eliminates the effect of extreme item - The median can be located graphically - It is affected more by fluctuation of sampling than the arithmetic mean. - It is not based on all the items of the series 3points *1 mark = 3 mark	3	
7	➤ Table Number ➤ Title of the table ➤ Head Note ➤ Caption ➤ Body of the table ➤ Source Note ➤ Foot Note 6 points *0.5=3 mark	3	
8	➤ It is easy to understand and simple to calculate ➤ It should be rigidly defined ➤ It should be based on all observations and it should not be affected by extreme observations ➤ It should be amenable to further algebraic treatment ➤ It should have sampling stability 6 points *0.5=3 mark	3	
9	Consumer price Index is designed to measure the change in the cost of living of workers because of change in the retail price. A change in the price level affects the cost of living of the people. People consume some different types of commodities. So there is need to construct consumer's price index. Consumer price index can be used in different places for many purposes. Proper explanation deserve 3 mark	3	
10	a) Aggregate of facts/data b) Numerically expressed c) Affected by different factors d) Collected or estimated e) Reasonable standard of accuracy f) Predetermined purpose g) Comparable h) Systematic collection 6 points *0.5=3 mark	3	
	<u>PART C</u>		15*4 =30
<u>III</u>	1. Direct Personal observation: This is a very general method of collecting primary data. Here the investigator directly contacts the informants, solicits their cooperation and enumerates the data. The information are collected by direct personal interviews.	7	

	2. Indirect Oral Interviews : This is an indirect method of collecting primary data. Here information are not collected directly from the source but by interviewing persons closely related with the problem. This method is applied to apprehend culprits in case of theft, murder etc. 3. Mailed Questionnaire method: This is a very commonly used method of collecting primary data. Here information are collected through a set of questionnaire. 4. Schedule Method: In case the informants are largely uneducated and non-responsive data cannot be collected by the mailed questionnaire method. 5. From Local Agents: Sometimes primary data are collected from local agents or correspondents. These agents are appointed by the sponsoring authorities (4 item*1.5=6,, overall presentation 1 mark.)		
<u>IV-</u>	1. Statistics laws are true on average. 2. Statistics are aggregates of facts, so a single observation is not a statistic. 3. Statistics deal with groups and aggregates only. 4. Statistical methods are best applicable to quantitative data. 5. Statistics cannot be applied to heterogeneous data. 6. If sufficient care is not exercised in collecting, analyzing and interpreting the data, statistical results might be misleading. 7. Only a person who has an expert knowledge of statistics can handle statistical data efficiently. 7 points *1=7	7	
<u>V</u>	Median class is the value of (N/2)th item (i.e.160/2) = 80th item of the series, which lies in 35–40 class interval. Median = $L + \frac{(N/2 - CF)}{F} * I$ $= 35 + \frac{80 - 75}{75 - 40} * 5$ $= 35 + \frac{5}{35} * 5$ $= 35 + 0.714 * 5$ $= 35 + 3.57$ $= 38.57$ Equation= 2 mark , steps = 3 mark, final answer =2 mark	7	
<u>VI</u>	Merits: - Based on all the items. - Well-defined and definite measure of dispersion. - Least affected by sample fluctuations. - Suitable for algebraic treatment. Demerits : - Standard Deviation is comparatively difficult to calculate. - Much importance is given to the extreme values. (merits 3.5 marks and demerits 3.5 marks)	7	

VII	$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$ R=0.077 Student may use short cut equation = 2 mark Column and steps= 3 marks Final answer =2 marks	7	
VIII	According to A. M. Tuttle, “<i>Analysis of co-variation of two or more variables is usually called correlation</i>”. Types of Correlation :Correlation can be of following types - (i) Positive and Negative Correlation :If changes in two connected series is in the same direction, (ii) Linear an Non-Linear :If the amount of change in one variable tends to bear constant ratio of change in the other variable, the correlation is said to be linear. (iii) Simple, Partial and Multiple Correlation :When only two variables are studied, it is called simple correlation. If the common effect of two or more independent variables on one dependent data series is studied, it is called multiple correlations. For example if the study of rain, soil, temperature on potato production per acre is studied then it is multiple correlation. On the other hand, in partial correlation we recognize more than two variables, but consider only two variables to be influencing each other, the effect of other influencing variables being kept constant. Meaning 1 mark, 3 types *2=6	7	
IX	$P o l = \frac{\sum p_1}{\sum p_0} \times 100 = \frac{550}{400} = 137.5$ Equation = 2 marks , Steps = 3 marks , Final answer=2 marks	7	
X	Index Numbers are specialized averages designed to measure the change in a group of related variables over a period of time. (i) Help in Framing Suitable Policies (ii) They Reveal Trends (iii) Helps in Comparative Study (iv) Important in Forecasting (v) Universal Application Meaning -2 mark, 5 points *1=5	7	
XI	Continuous Series $\sum X / \sum N$ Mean =33.71 Equation -2 mark, steps-3 mark, final answer -2 mark.	7	

XII	X	IDI = X-X	7	
	100	100		
	150	50		
	200	0		
	250	50		
	300	100		
	Σx= 1000	ΣI DI		
	Mean = Σx / N = 1000/5 = 200			
	<u>ΣI DI</u>			
N				
M.D = ----- = 300 / 5 =60				
60/200=0.30				
Equation 1 mark, steps and column presentation-4 mark, final answer-2 mark				
XIII	Price Index =P01=P1/P0X100		7	
	1980	100.00		
	1981	113.33		
	1982	106.66		
	1983	120.00		
	1984	146.66		
	1985	133.33		
	Equation-2 mark, column wise preparation -4 mark, correct answer-1 mark			
XIV	(i) Time Reversal Test :Time reversal test is a test to determine whether a given method will work both ways in time, forward and back ward. This test means that if the index number of the current year is computed on the basis of base year (P01) and again the index number of the base year is calculated on the basis of current year (P10), both the index numbers should be reciprocal of each other i.e. product of both of them should be one. Symbolically, the following relation should be satisfied - P01 X P10 = 1		7	
	(ii) Factor Reversal Test :According to this test, an index of price, if multiplied by an index of quantity, with the same base, given years and commodities, should be equal to the true value ratio, i.e. equal to the ratio of current year price to total price of the base year. P01 XQ01 = ΣP1q1 ΣP0q0			
	If the product is not equal to the value ratio, there is, with reference to this test, an error in one or both of the index numbers. (3.5 mark for each test.)			